

Introduction

IGBT (Insulated Gate Bipolar Transistor) is a semiconductor device commonly used in power electronics.

It combines the characteristics of a MOSFET and a bipolar transistor, making it ideal for high-voltage and high-current applications.

It is primarily used in high-power applications, such as motor drives, power inverters, and renewable energy systems.

Structure:

The IGBT consists of a layered structure with a combination of MOSFET and BJT properties.

IGBT is a three-terminal device with Gate (G), Collector (C), and Emitter (E).

It has three terminals:

1. Collector (C) – connected to the positive side of the power supply.
2. Emitter (E) – connected to the negative or ground side.
3. Gate (G) – the control terminal that modulates the current between the collector and emitter.

It has four layers in its internal structure: N⁺ (Emitter), P (Base), N (Drift), and P⁺ (Collector).

- It has an N-channel MOSFET connected to a PNP transistor, which allows the IGBT to have fast switching capability with high current handling.
- The gate is insulated from the rest of the device by a thin oxide layer (similar to a MOSFET), allowing voltage to control the flow of current.

Working

IGBT operates in two distinct states: ON and OFF, controlled by the voltage applied to the gate.

1. OFF State (Blocking Mode): When there is no voltage or a negative voltage applied to the Gate with respect to the Emitter ($V_{ge} < \text{Threshold Voltage}$), the device is in the OFF state.
 - The gate is not conducting, N-channel MOSFET remains inactive
 - The P-N junction between the N-base and P-base regions is reverse-biased, and no current flows.
2. ON State (Conduction Mode): When a positive voltage is applied between the Gate and Emitter ($V_{ge} > \text{Threshold Voltage}$), the IGBT turns ON.
 - The positive gate voltage creates an electric field that forms an inversion layer (N-channel) in the P-region beneath the gate.
 - This allows electrons to flow from the Emitter into the N-base, creating a current path.

- As the gate voltage is applied, it switches ON the MOSFET portion of the IGBT, and once conduction starts, the BJT portion of the device takes over, allowing a large current to flow from the Collector to Emitter.
- The IGBT conducts heavily with minimal voltage drop

Advantages

- High Power Efficiency: IGBTs reduce power loss in high-voltage applications, making them very energy-efficient.
- Fast Switching Speed: IGBTs can switch quickly between ON and OFF states, allowing them to handle rapid changes in current, which is essential in applications like motor drives and power inverters.
- Compact Design: They offer a compact and lightweight design, ideal for modern electronic devices and systems that need to save space.
- Thermal Stability: IGBTs are designed to handle high temperatures, making them reliable for long-term use in industrial environments.

Applications

1. Industrial Motor Drives: Used for speed control in industrial motors due to its high-power efficiency.
2. Electric and Hybrid Vehicles: Critical in electric drivetrains for efficient power conversion and control.
3. Renewable Energy Systems: Widely used in solar inverters and wind power systems to convert and control power.
4. Uninterruptible Power Supplies (UPS): Ensures consistent power supply, essential in backup systems.
5. Consumer Electronics: Used in appliances that require power control, such as induction cookers and microwave ovens.
6. Solid-State Drives (SSD): Employed in SSD controllers for efficient data storage and retrieval due to its fast switching ability.

